Projekt 2

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Zadanie

- Rozpoznávanie gest ruky z RGB obrazu – 6 skupín (jednotlivé gestá). Aplikovať CNN na kameru a urobiť testovanie CNN na jednotlivé gestá
- Triedy - “0”, “1”, “2”, “3”, “4”, “5”, “6”

- Rozpoznávanie gest ruky z hĺbkovej mapy – 6 skupín (jednotlivé gestá)
- Triedy - “bye”, “c”, “dunno”, “pray”, “rock”, “up”

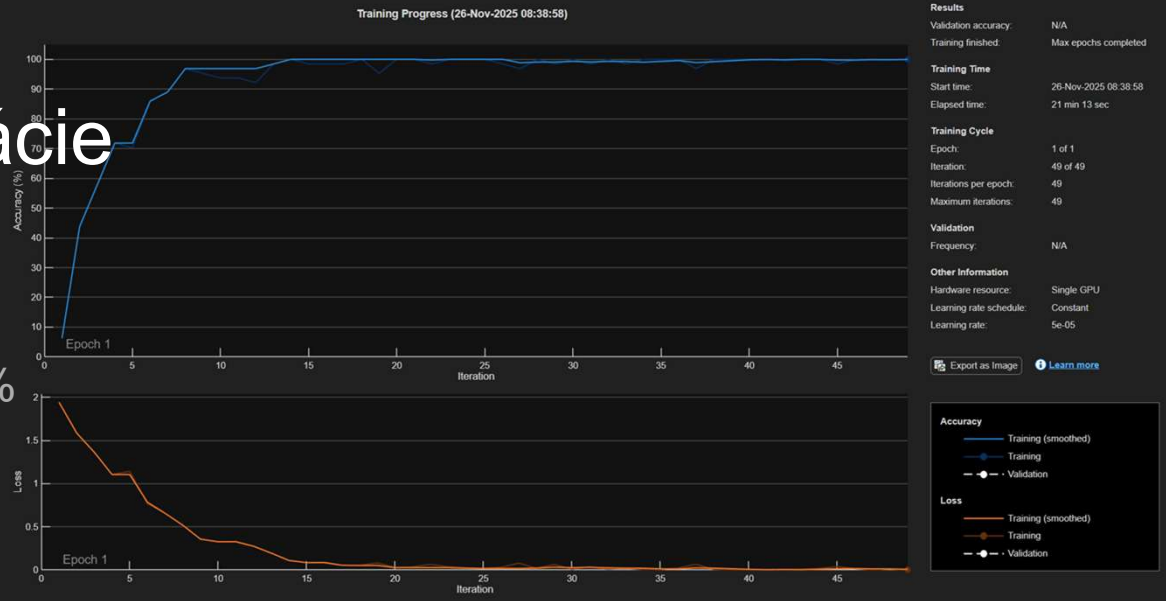
Inception v3 - bez augmentácie

- Parametre trénovania
  - Mini batch size - 64
  - Initial learn rate - 0.00005
  - Shuffle - every epoch

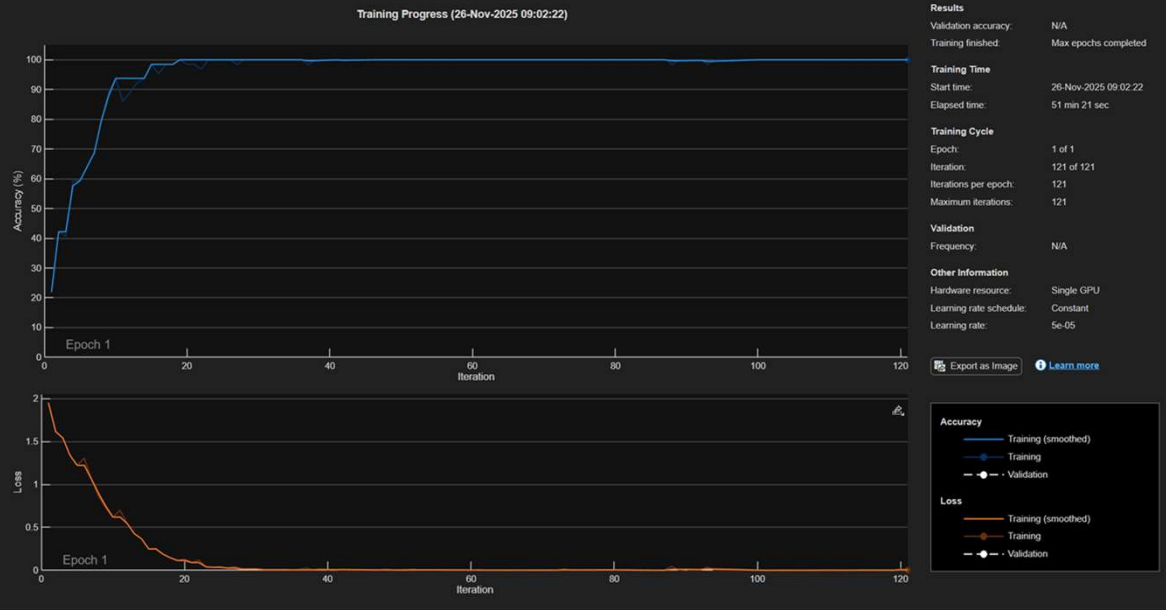
Inception v3 - bez augmentácie



99,87%

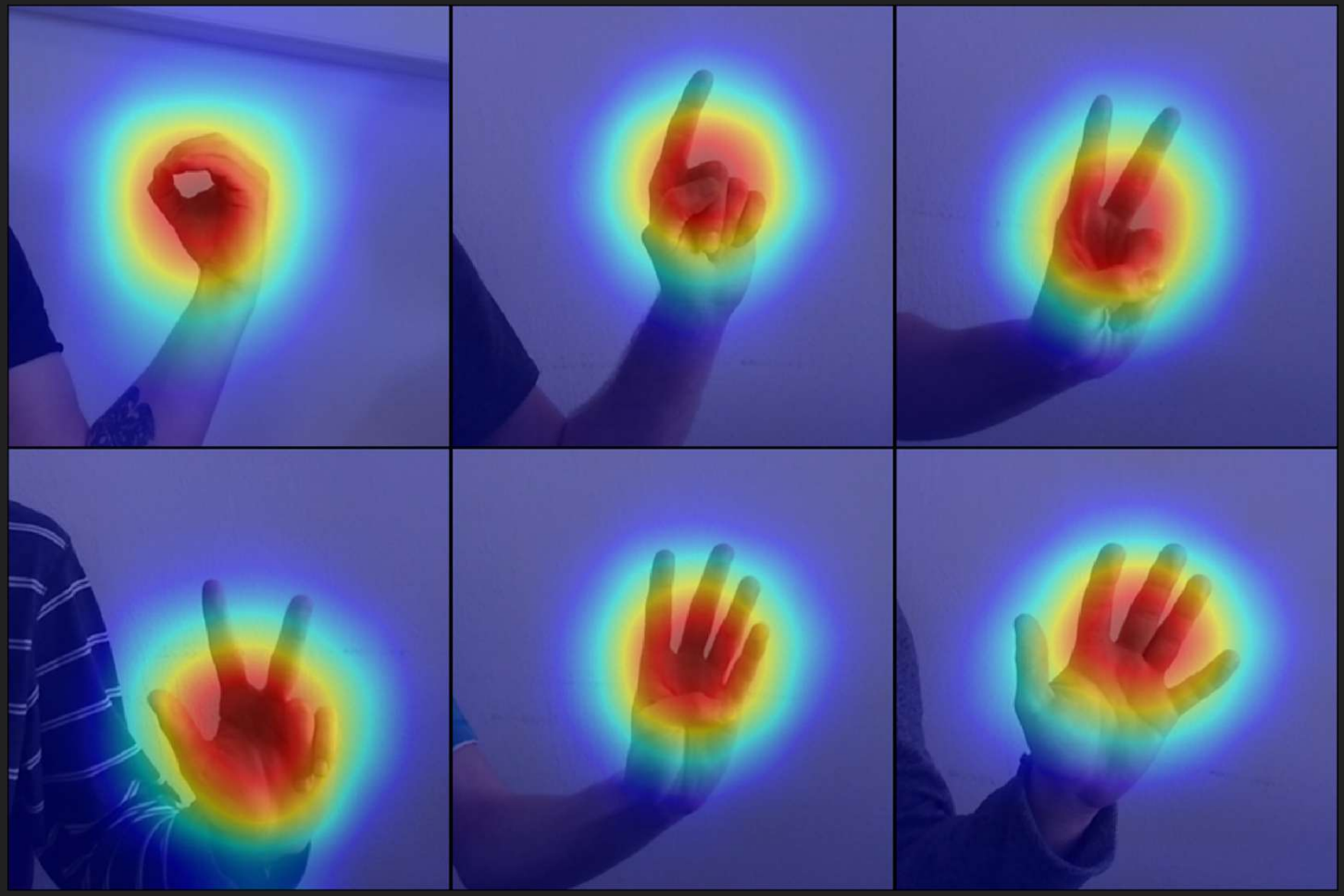


100%

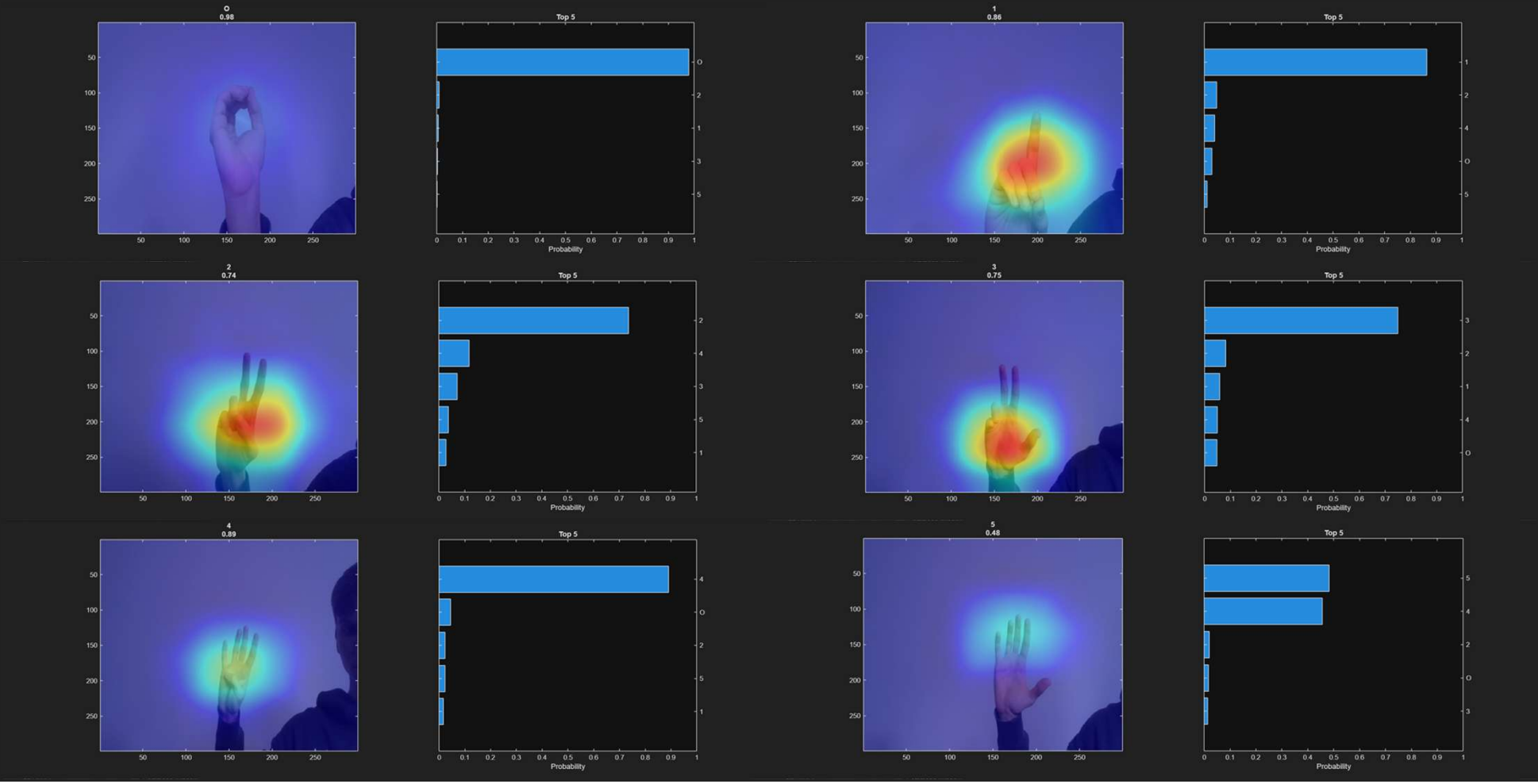


Inception v3 - bez augmentácie

- GradCAM



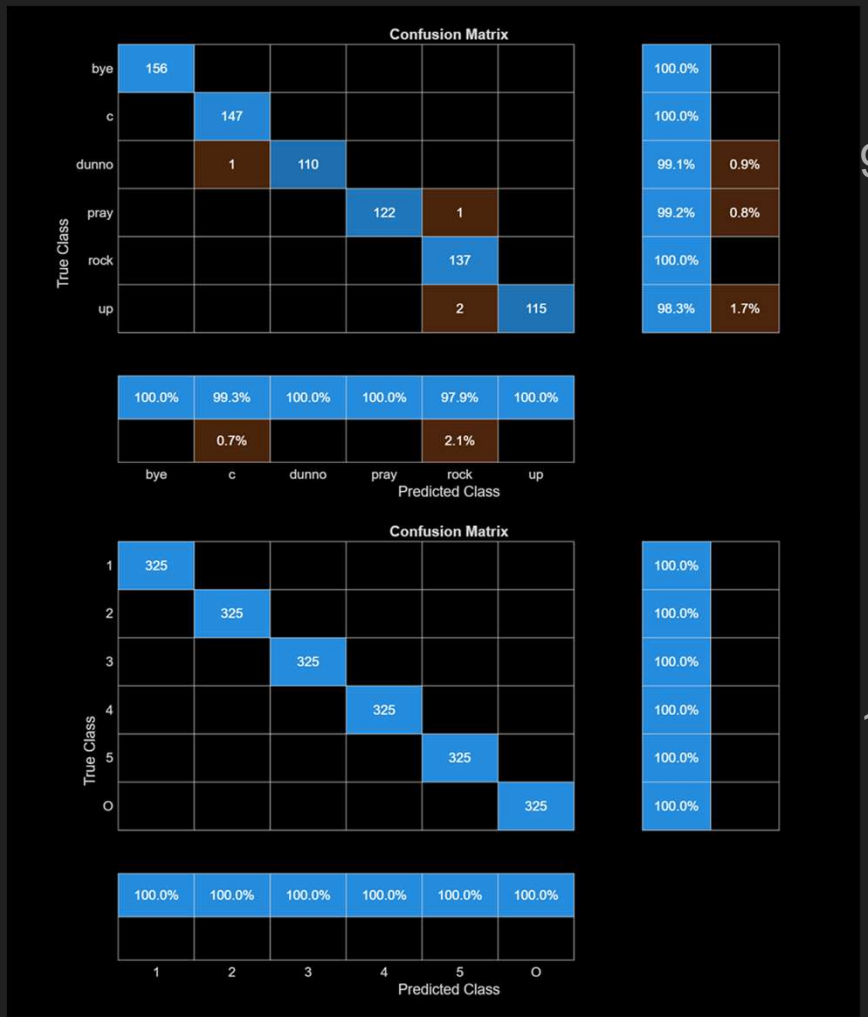
Inception v3 - bez augmentácie



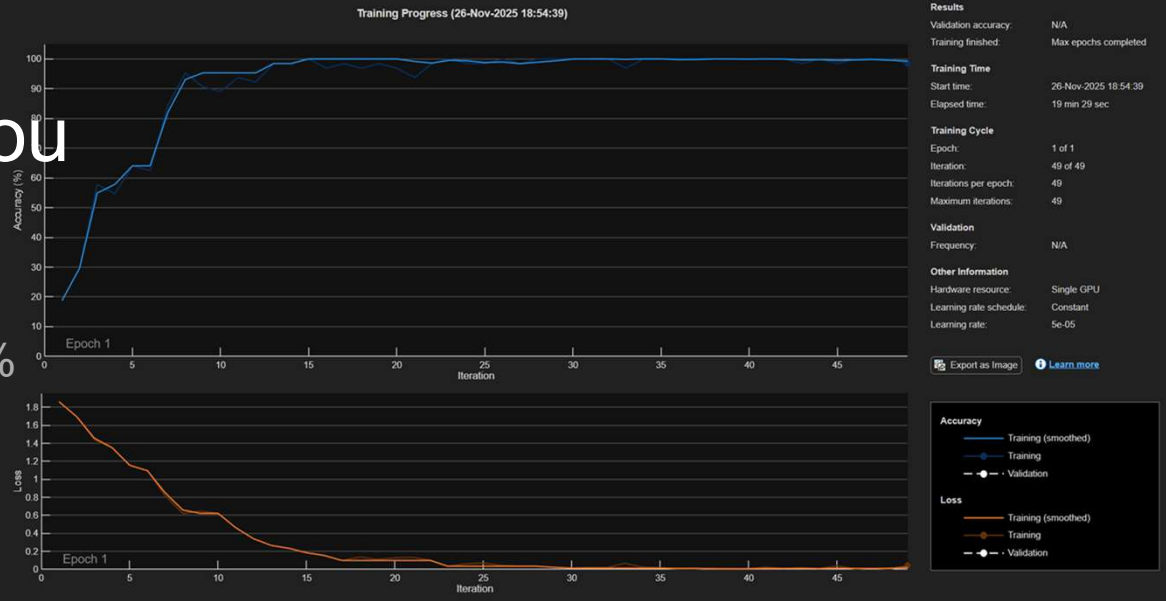
Inception v3 - s augmentáciou

- Augmentácia
  - Rotácia - [-15; 15] °
  - Translácia - [-15; 15] px
  - Prevrátenie v osi X
  - Priblíženie / oddialenie [0.8; 1.2]

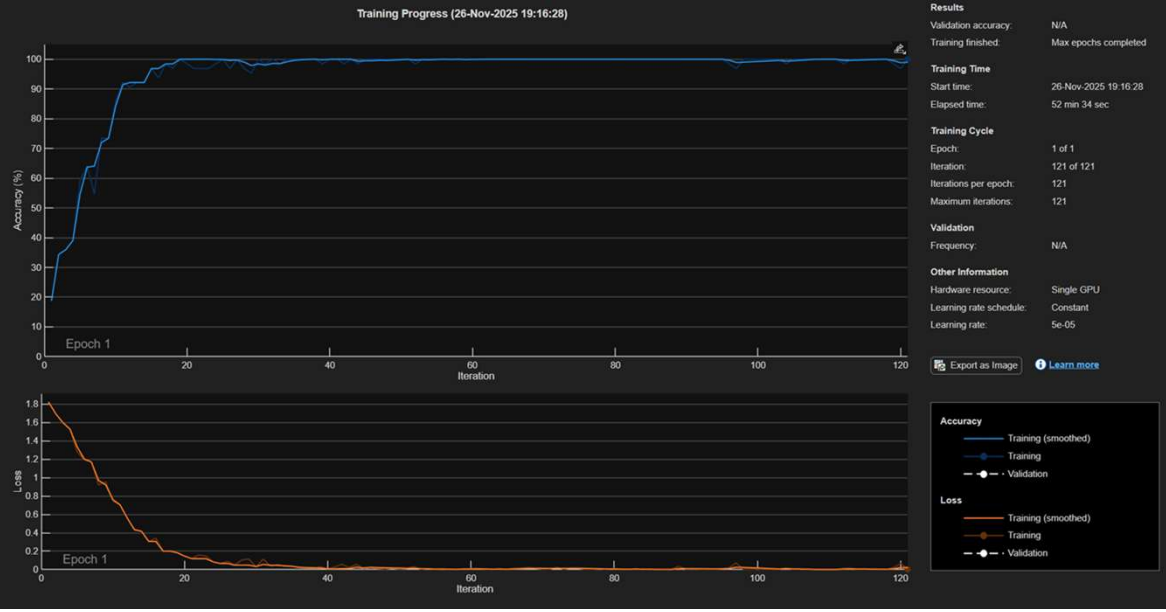
Inception v3 - s augmentáciou



99,49%

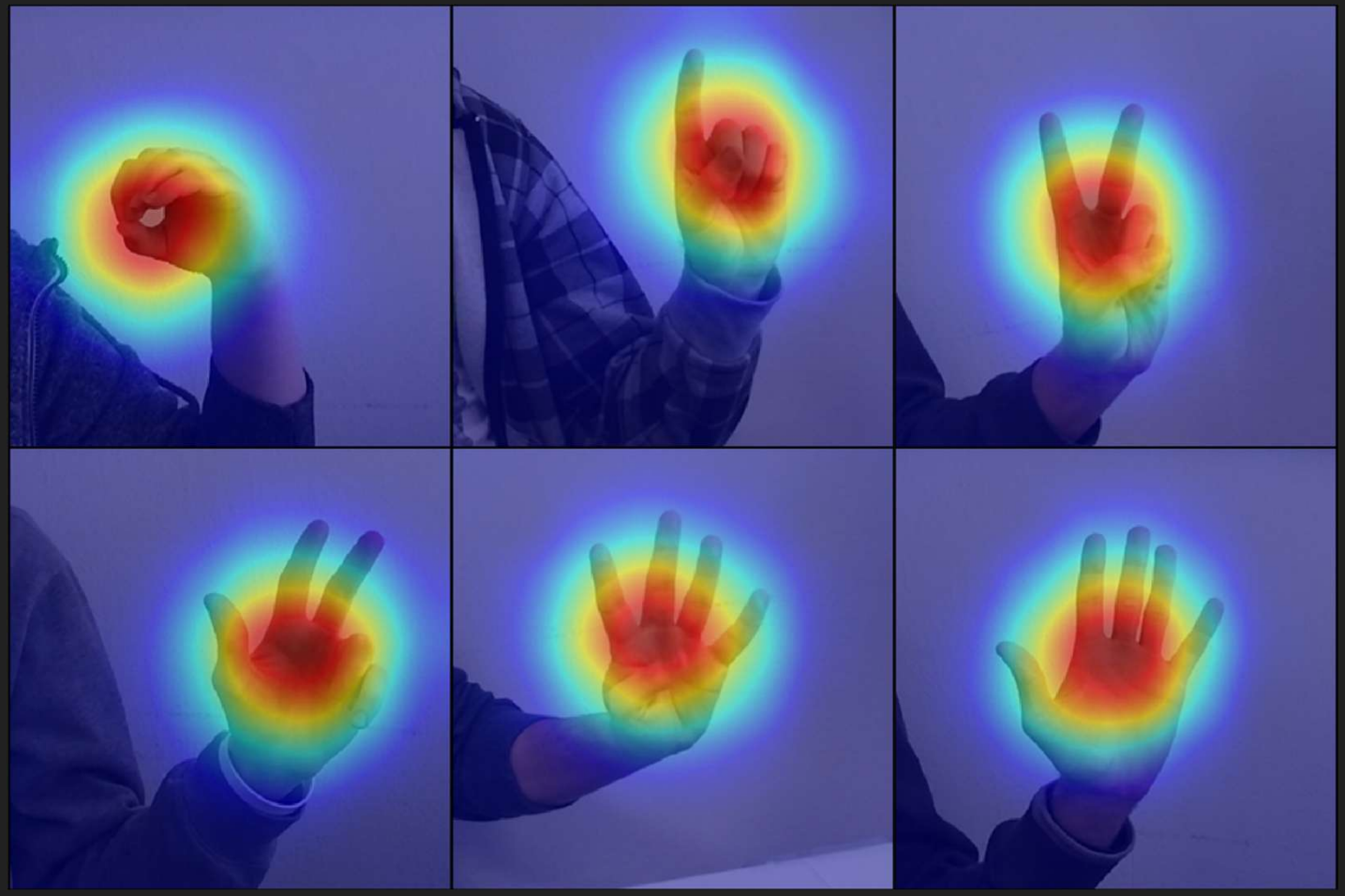


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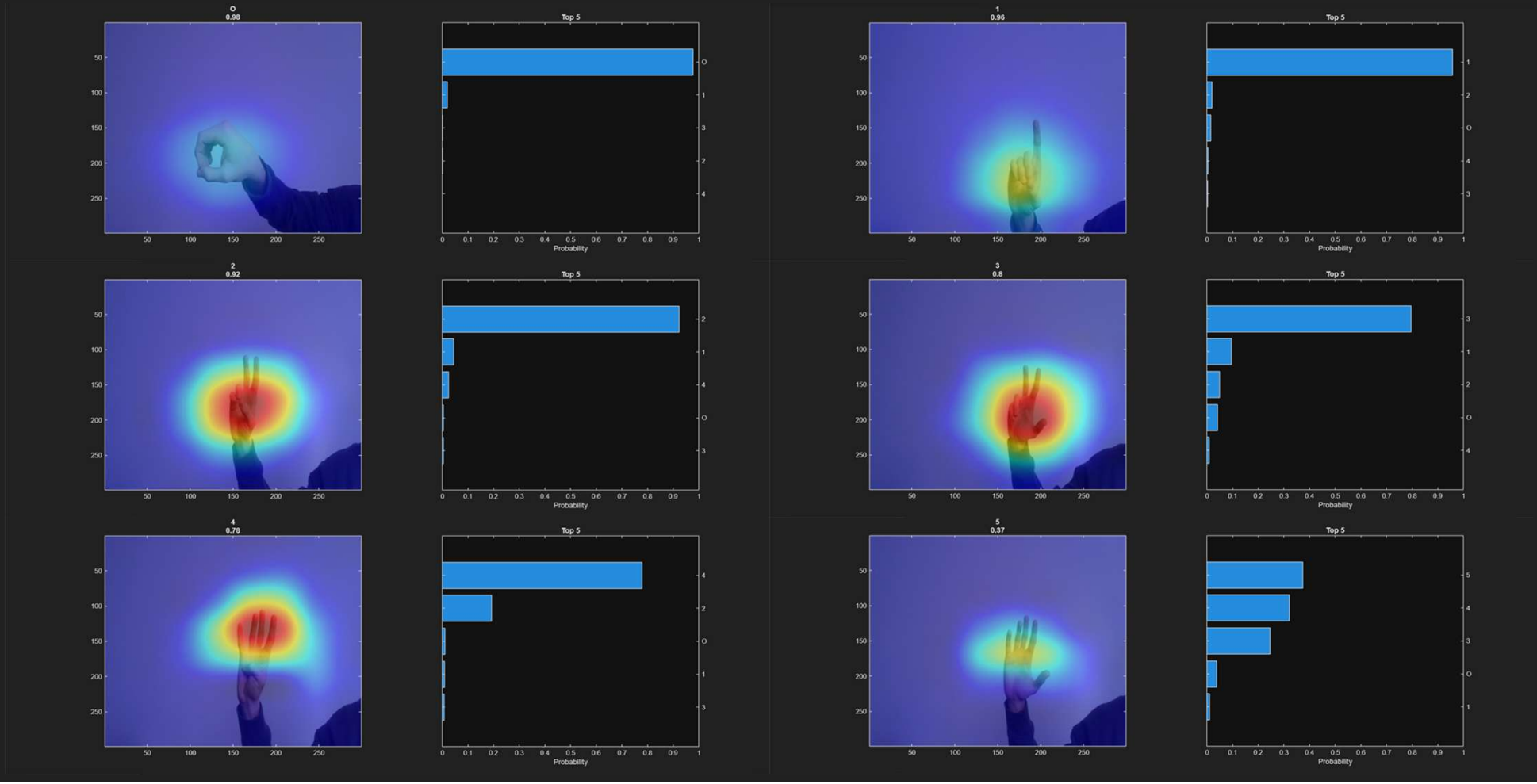


Inception v3 - s augmentáciou

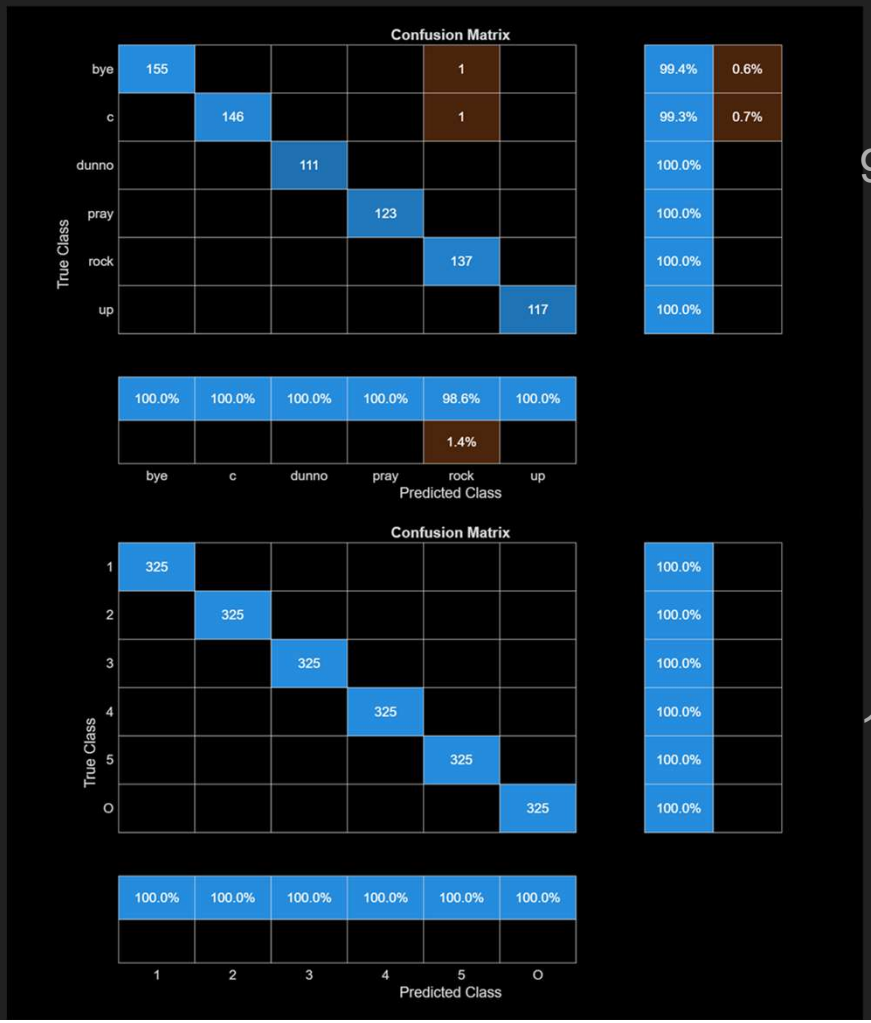
- GradCAM



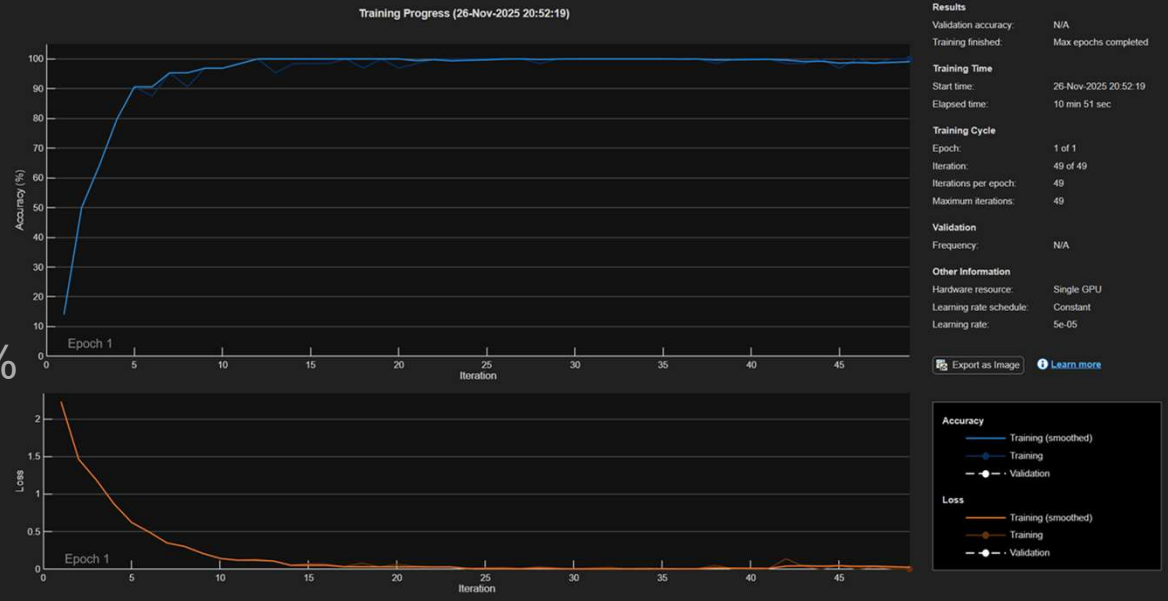
Inception v3 - s augmentáciou



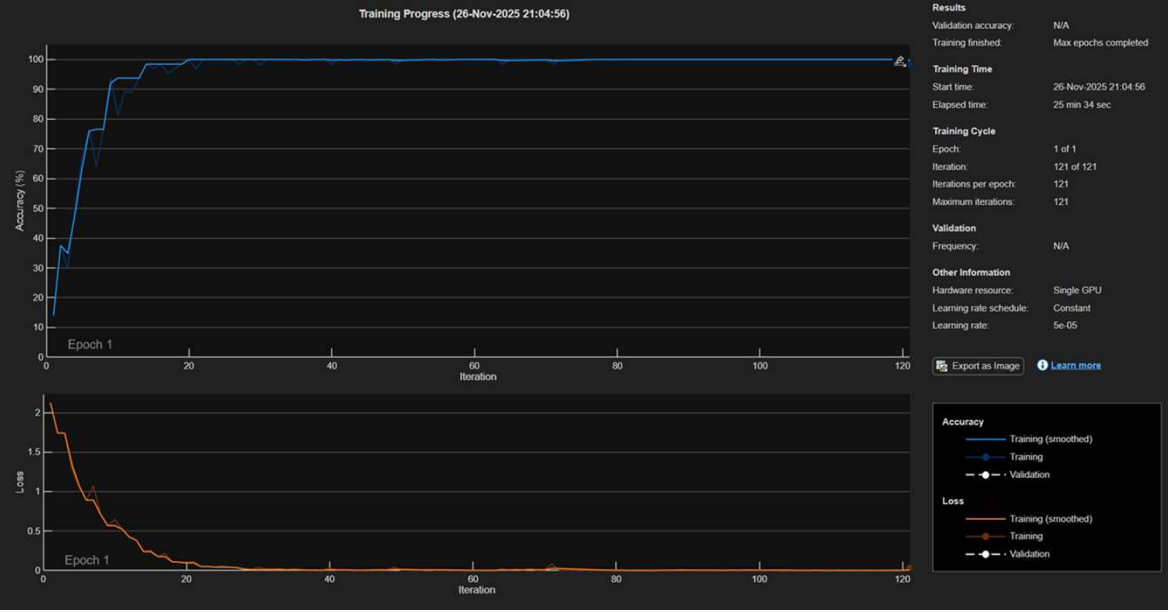
ResNet 50



99,75%

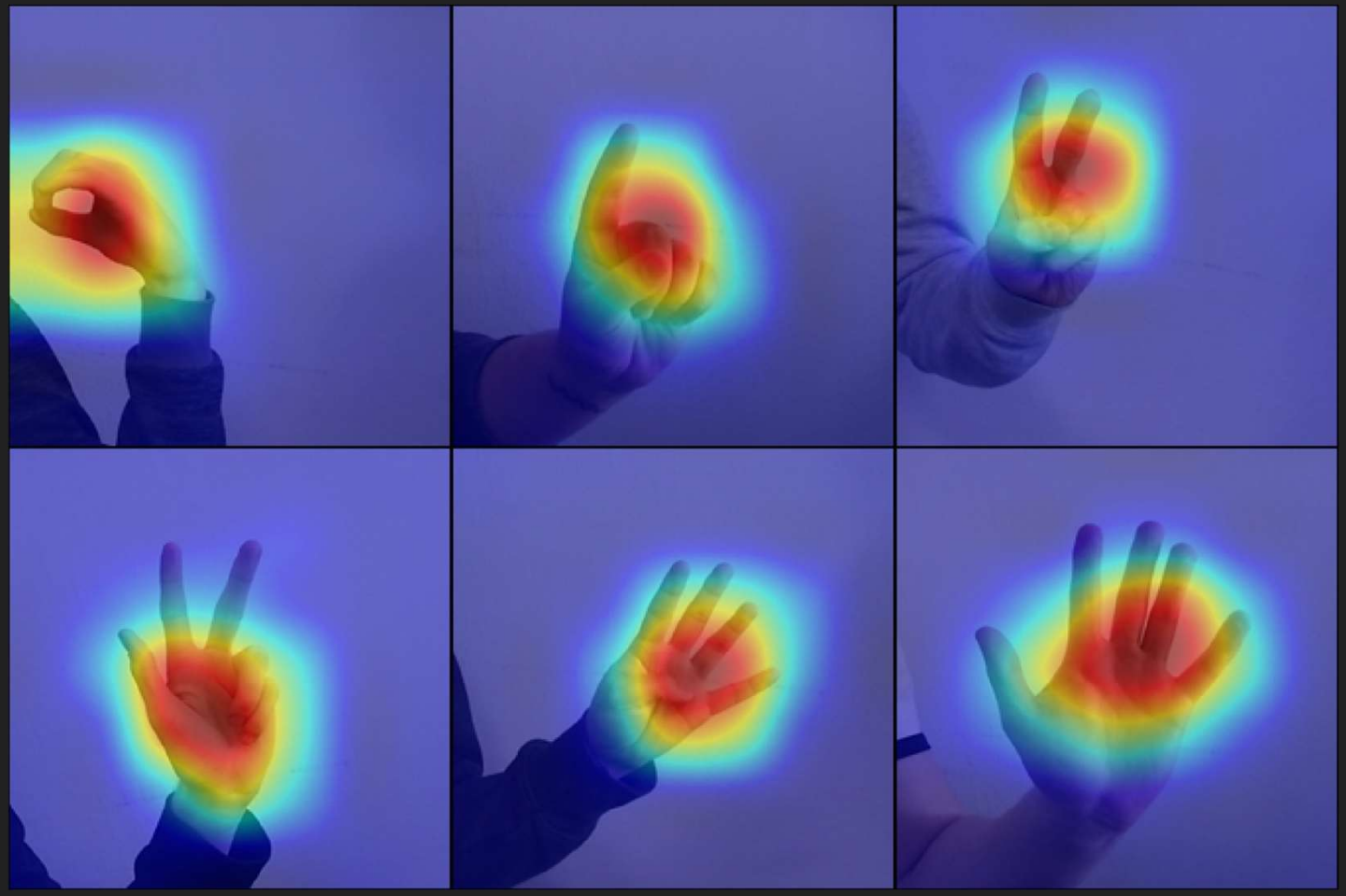


100%

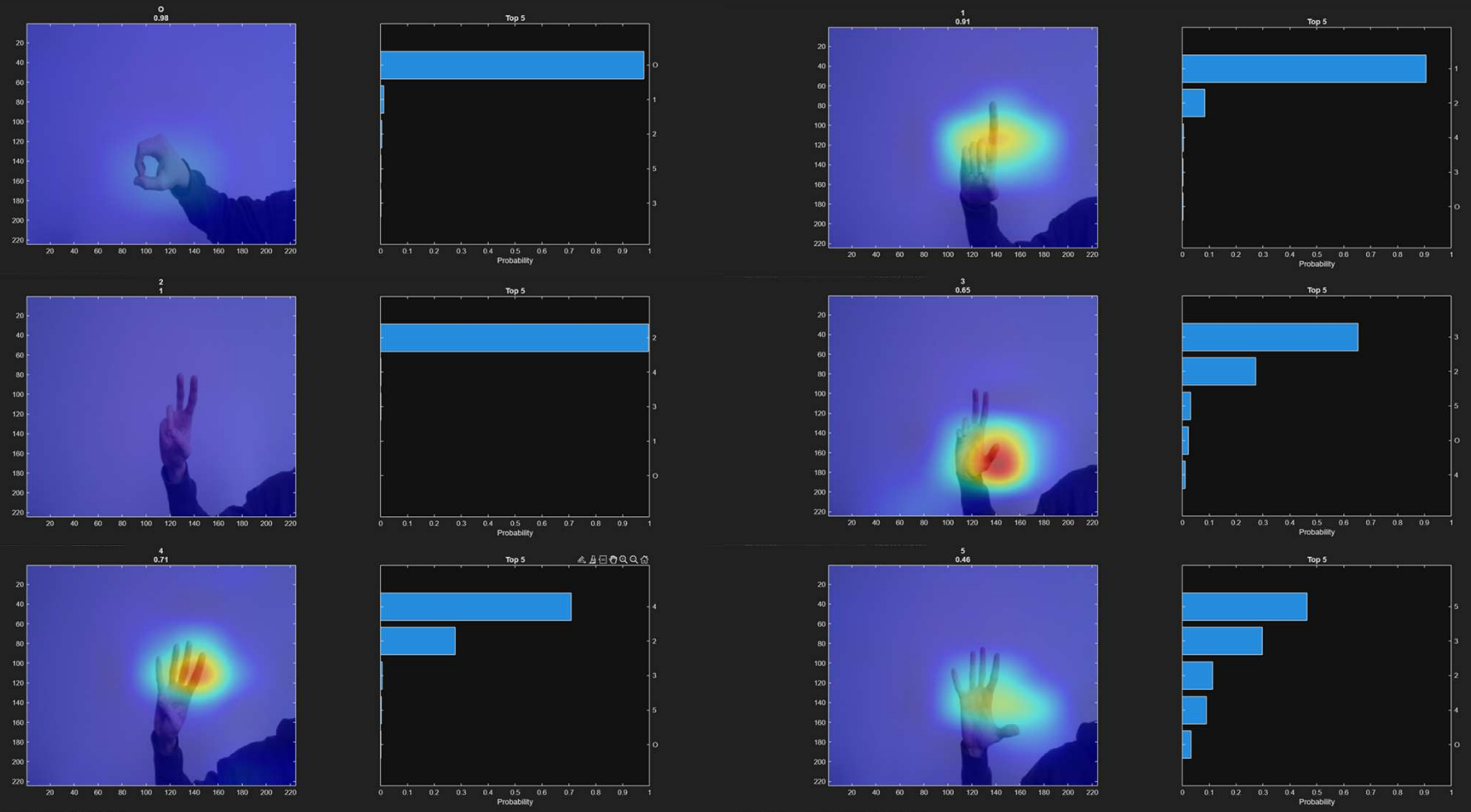


ResNet 50

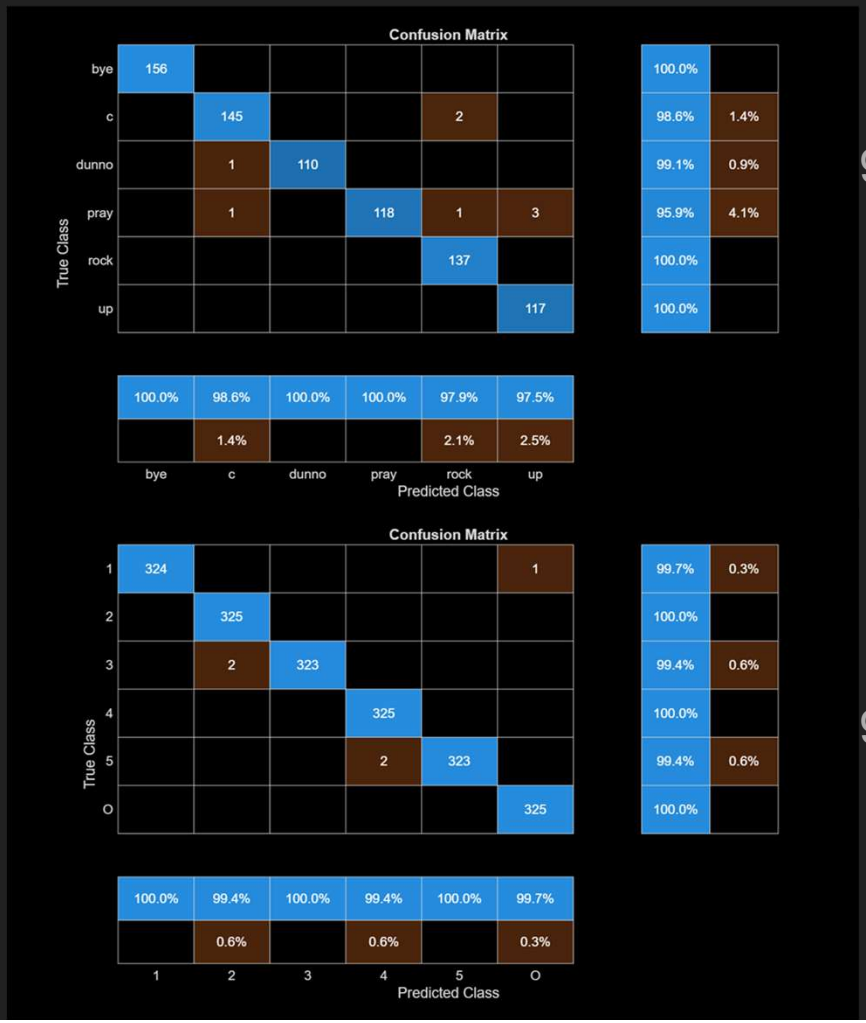
- GradCAM



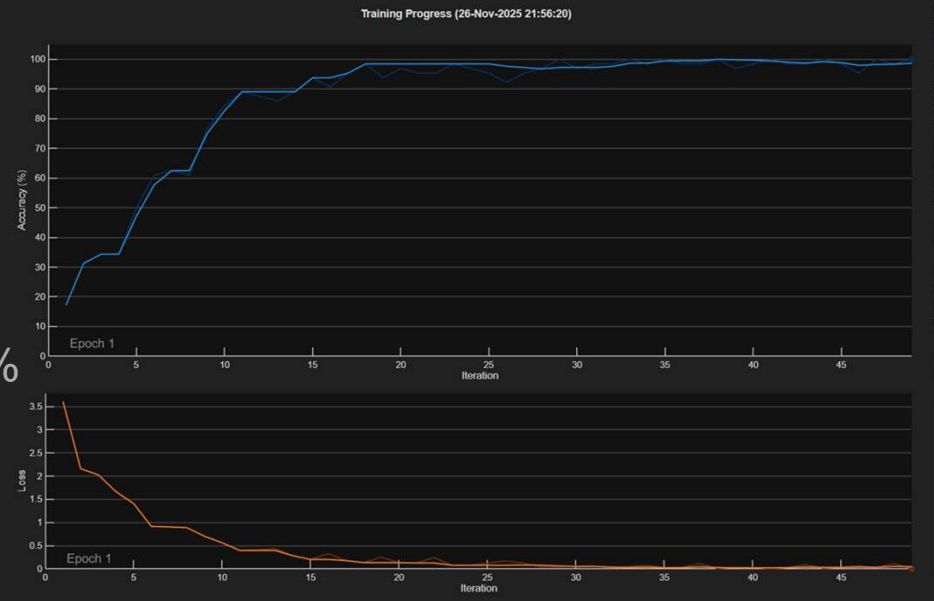
ResNet 50


O
0.98
Top 5
Probability
1
0.91
Top 5
Probability
2
1
Top 5
Probability
3
0.65
Top 5
Probability
4
0.71
Top 5
Probability
5
0.46
Top 5
Probability

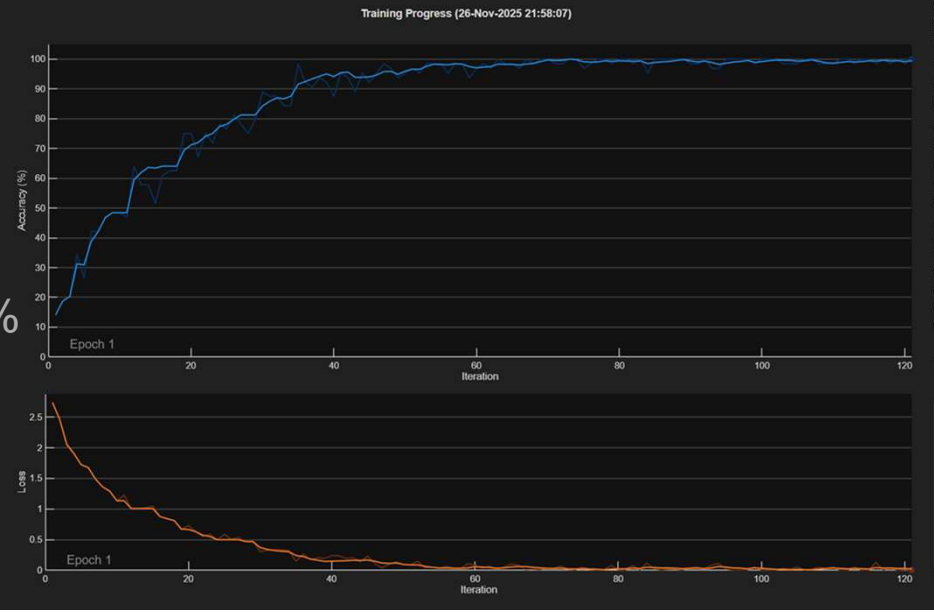
GoogleNet



98,99%

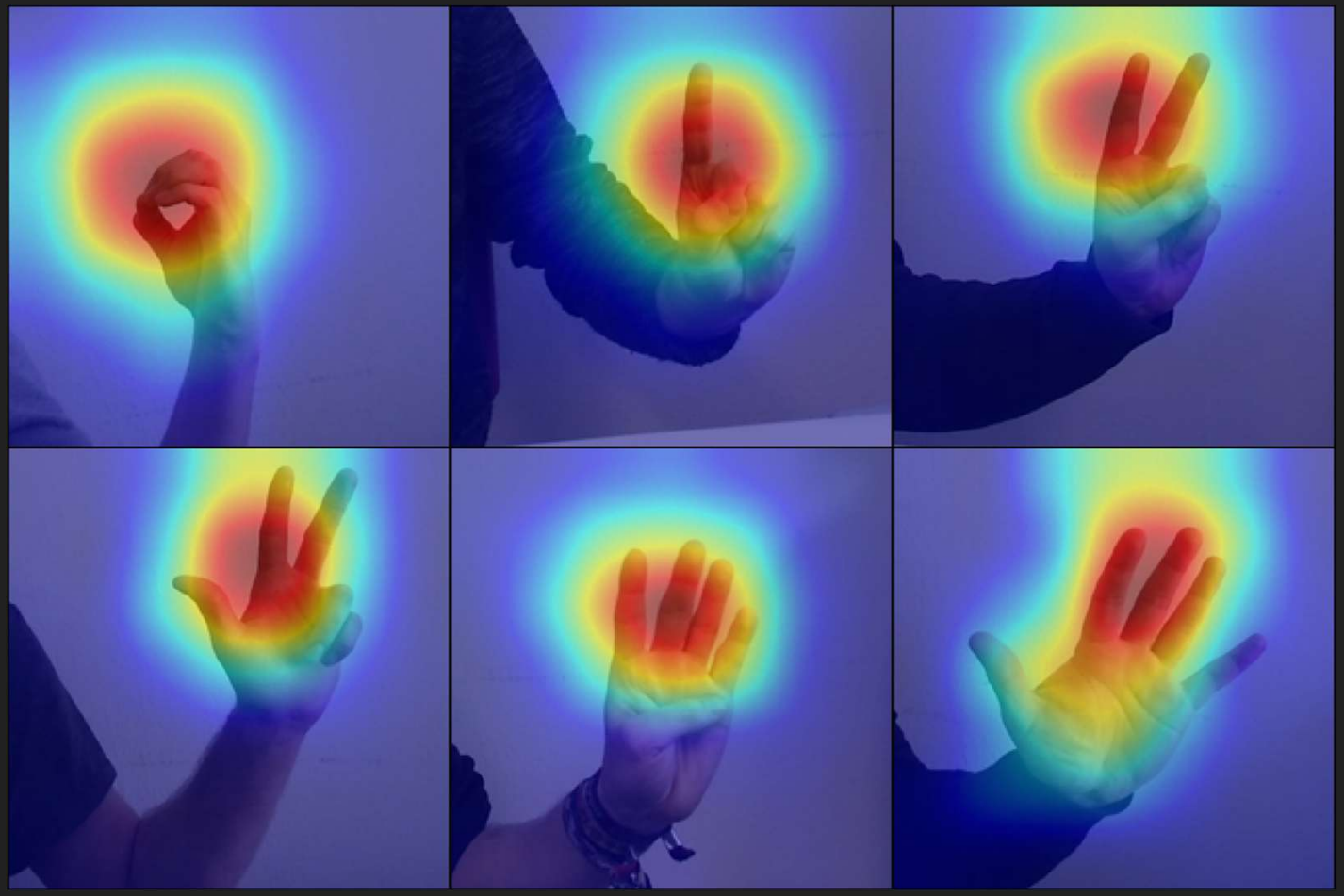


99,74%

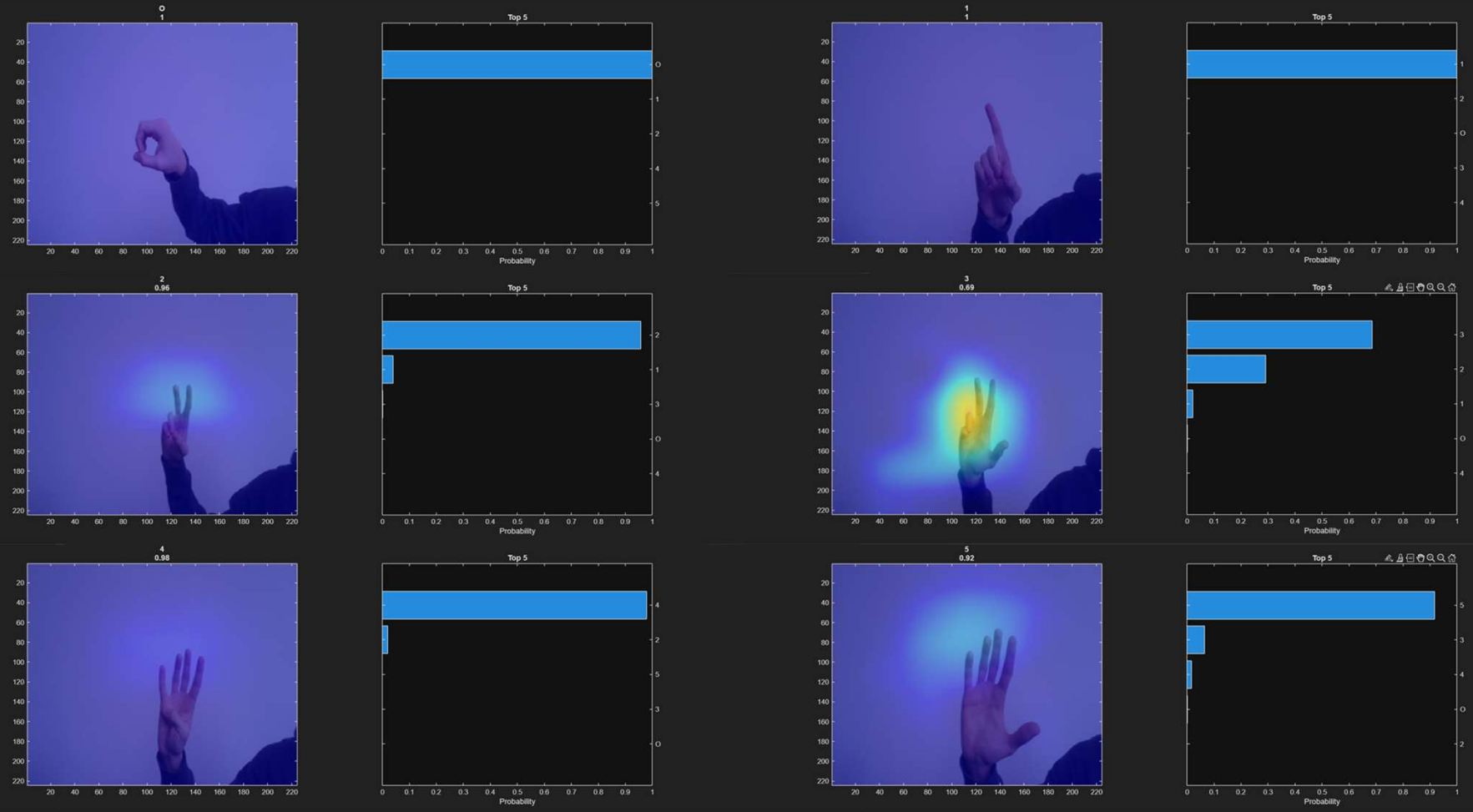


GoogleNet

- GradCAM



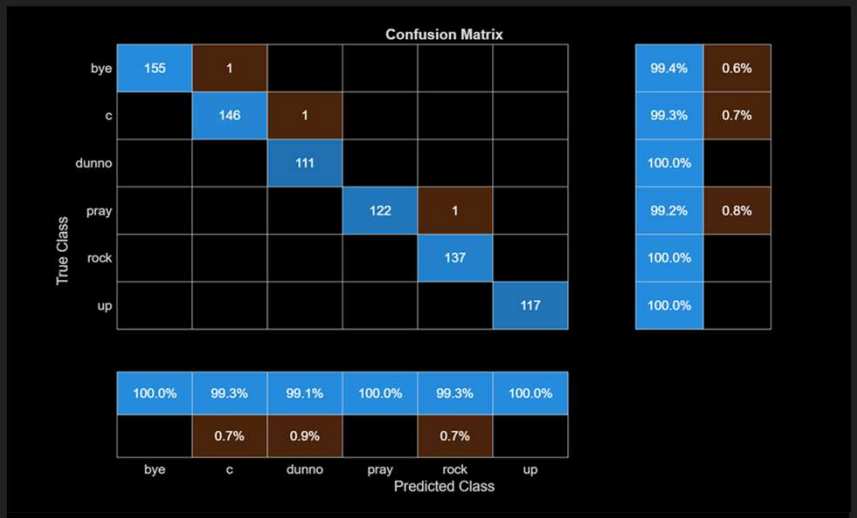
GoogleNet

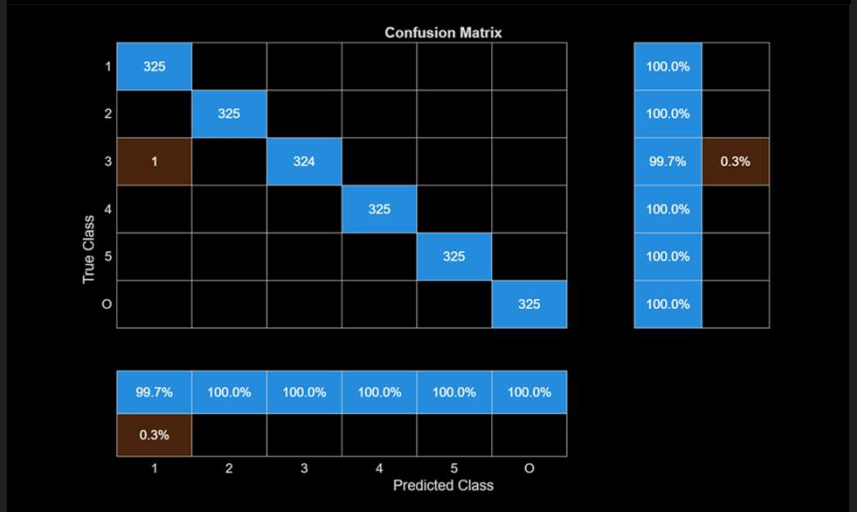

0
1
Top 5
Probability
1
1
Top 5
Probability
2
0.96
Top 5
Probability
3
0.69
Top 5
Probability
4
0.98
Top 5
Probability
5
0.92
Top 5
Probability

GoogleNet - upravené parametre trénovania

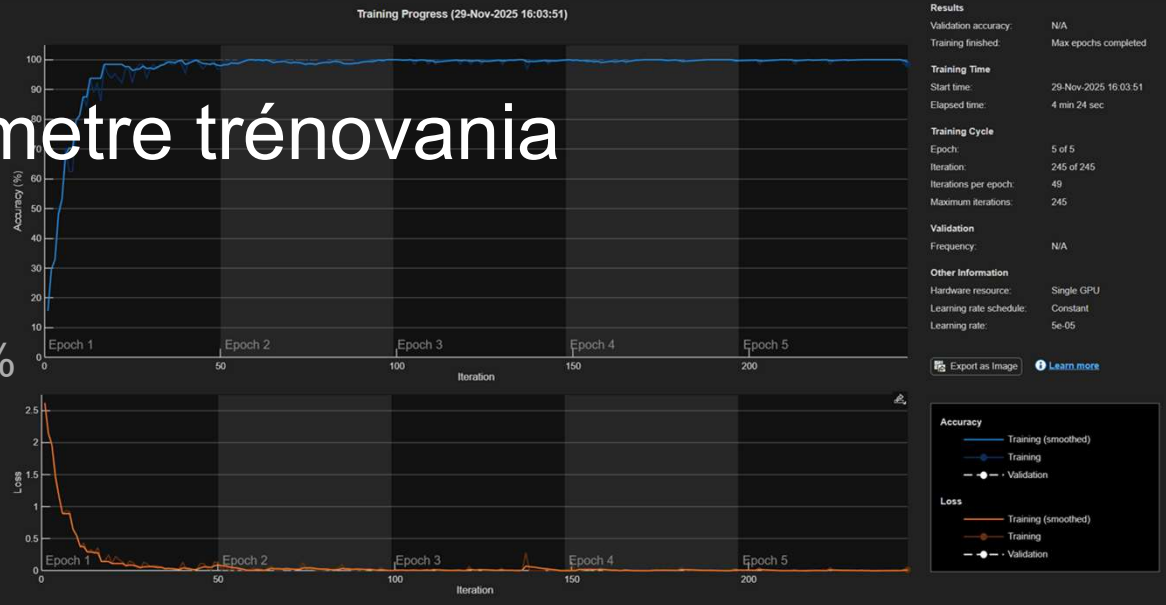
- Pridané parametre trénovania
  - L2 Regularization - 0.001
  - Learn rate drop factor - 0.1
  - Learn rate drop period - 1

GoogleNet - upravené parametre trénovania

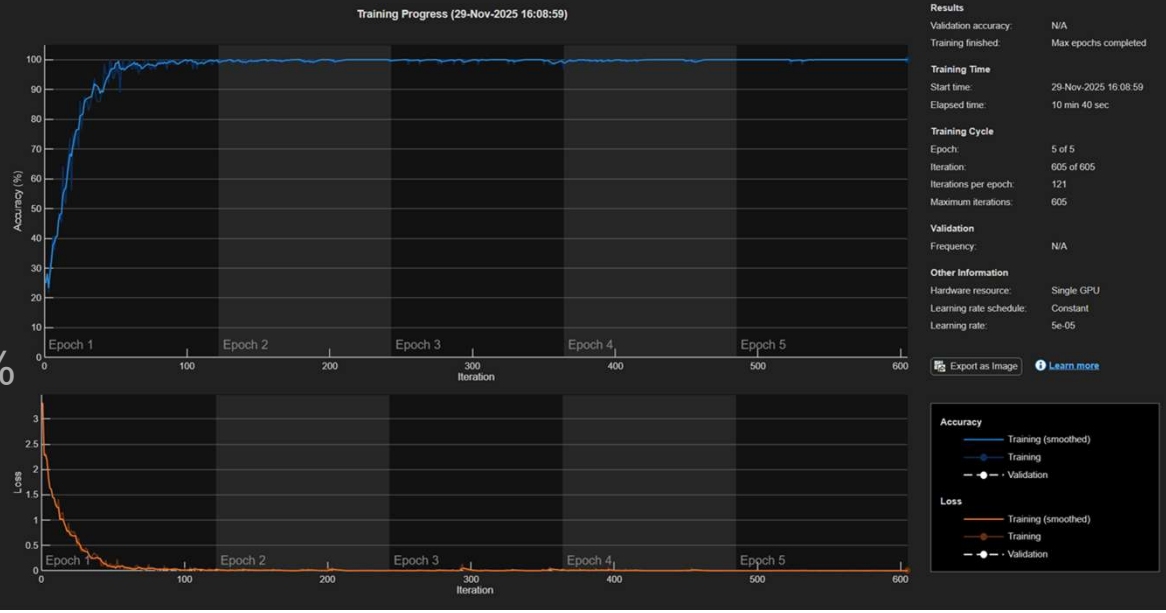




99,62%

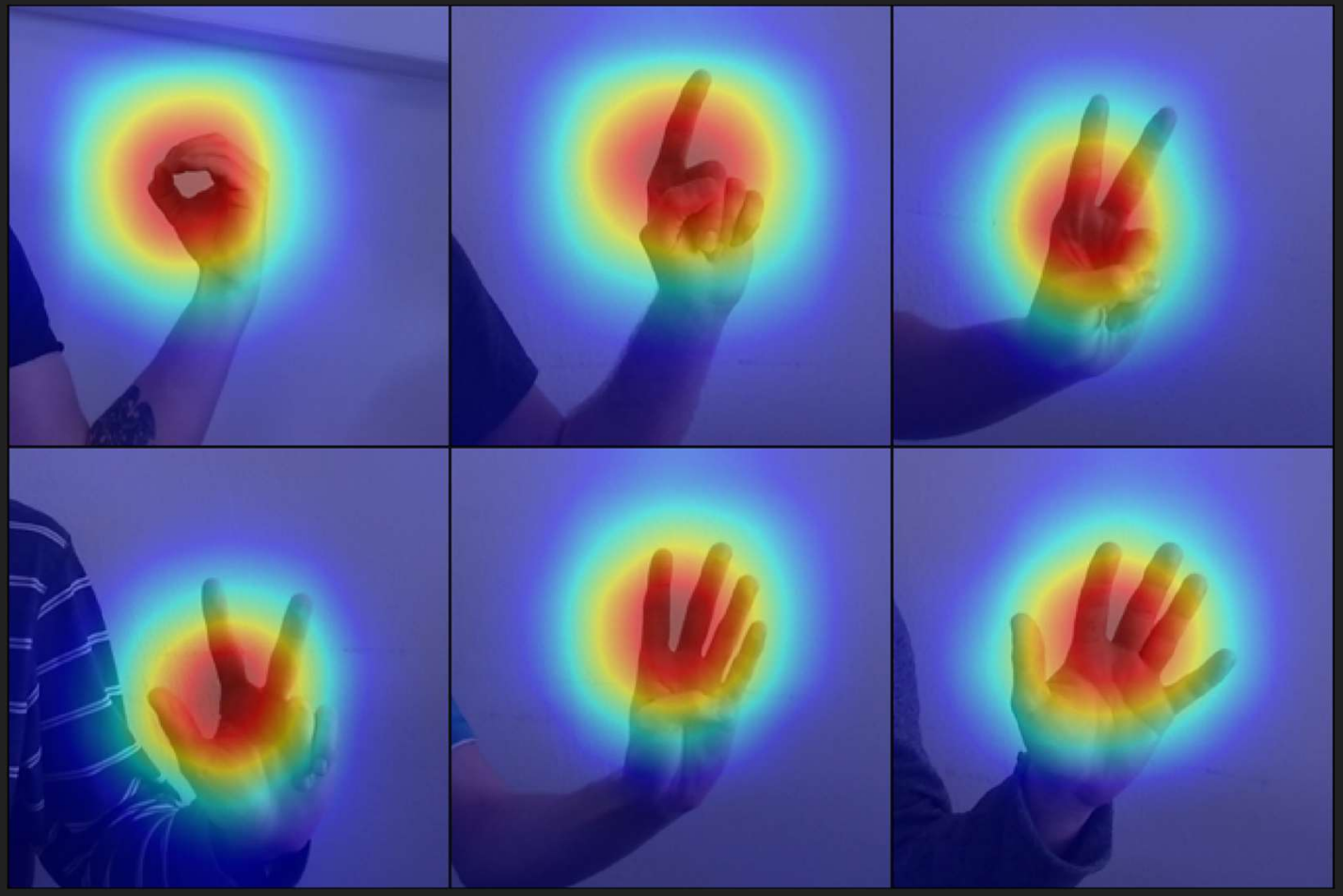


99,95%



GoogleNet - upravené parametre trénovania

- GradCAM



GoogleNet - upravené parametre trénovania

